

ChIP-seq for Histone modifications, Transcription Factor Binding Sites, ChromHMM, DNase-seq Hypersensitivity And ATAC-seq Analysis

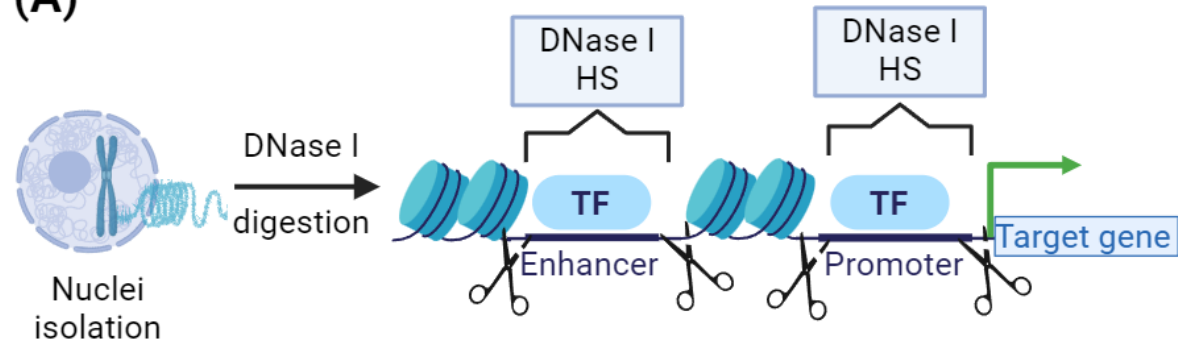
April 18 2026
Phuc-Loi Luu, PhD

Content

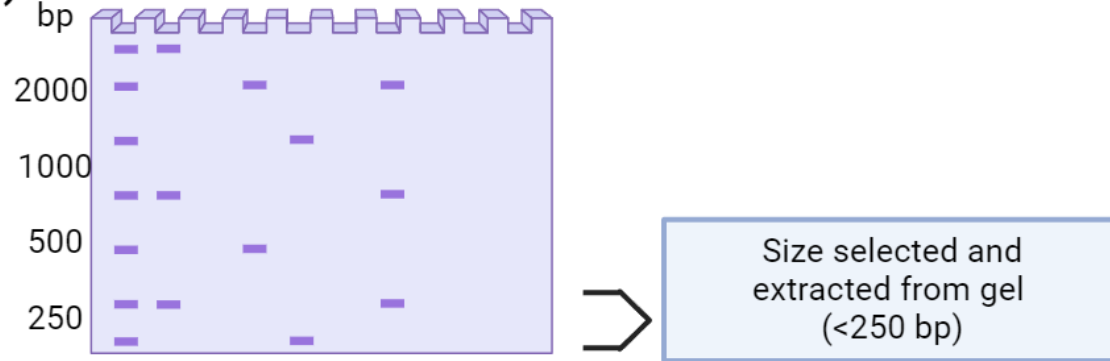
- 1. Dnase-seq for Chromatin Accessibility and TF footprints**
- 2. ATAC-seq**
- 3. ChIP-seq for Histone modifications and TFBSs & ChromHMM**
- 4. Reading the output of ChIP-Seq and ATAC-seq**

Dnase-seq for Chromatin Accessibility and TF footprints

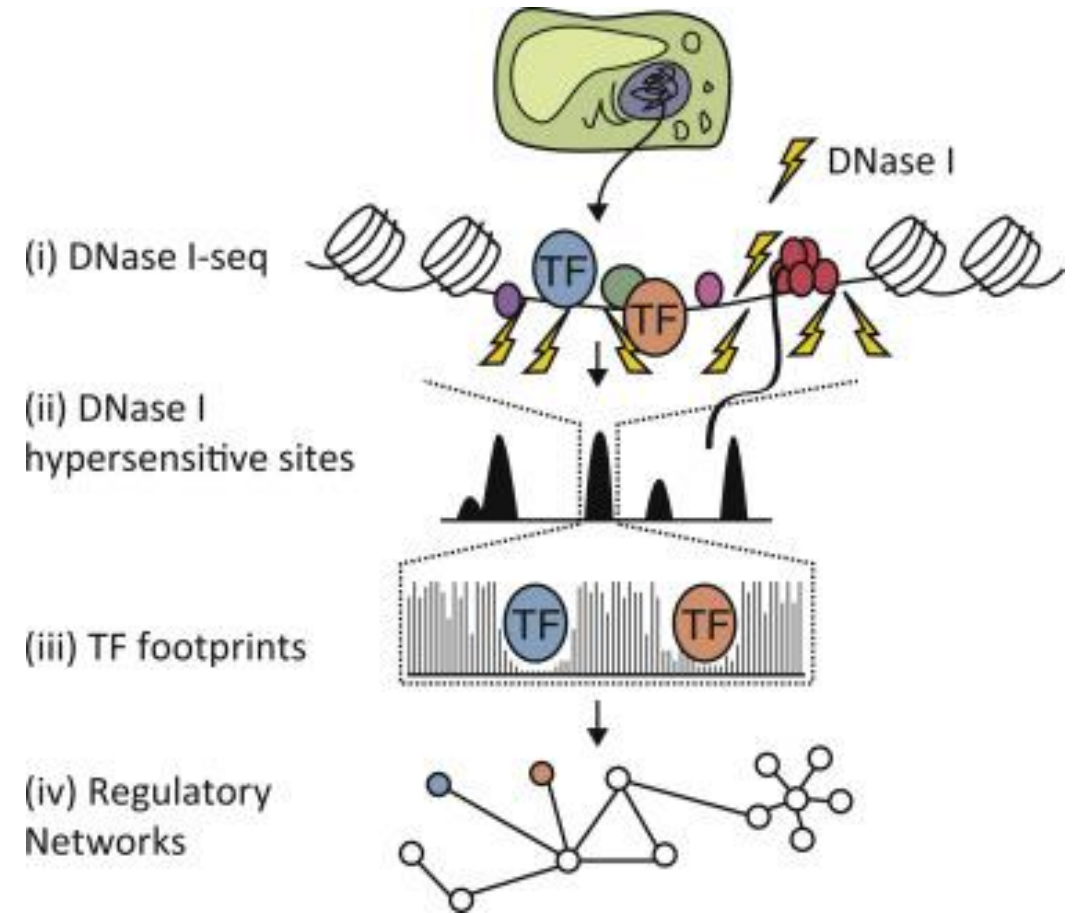
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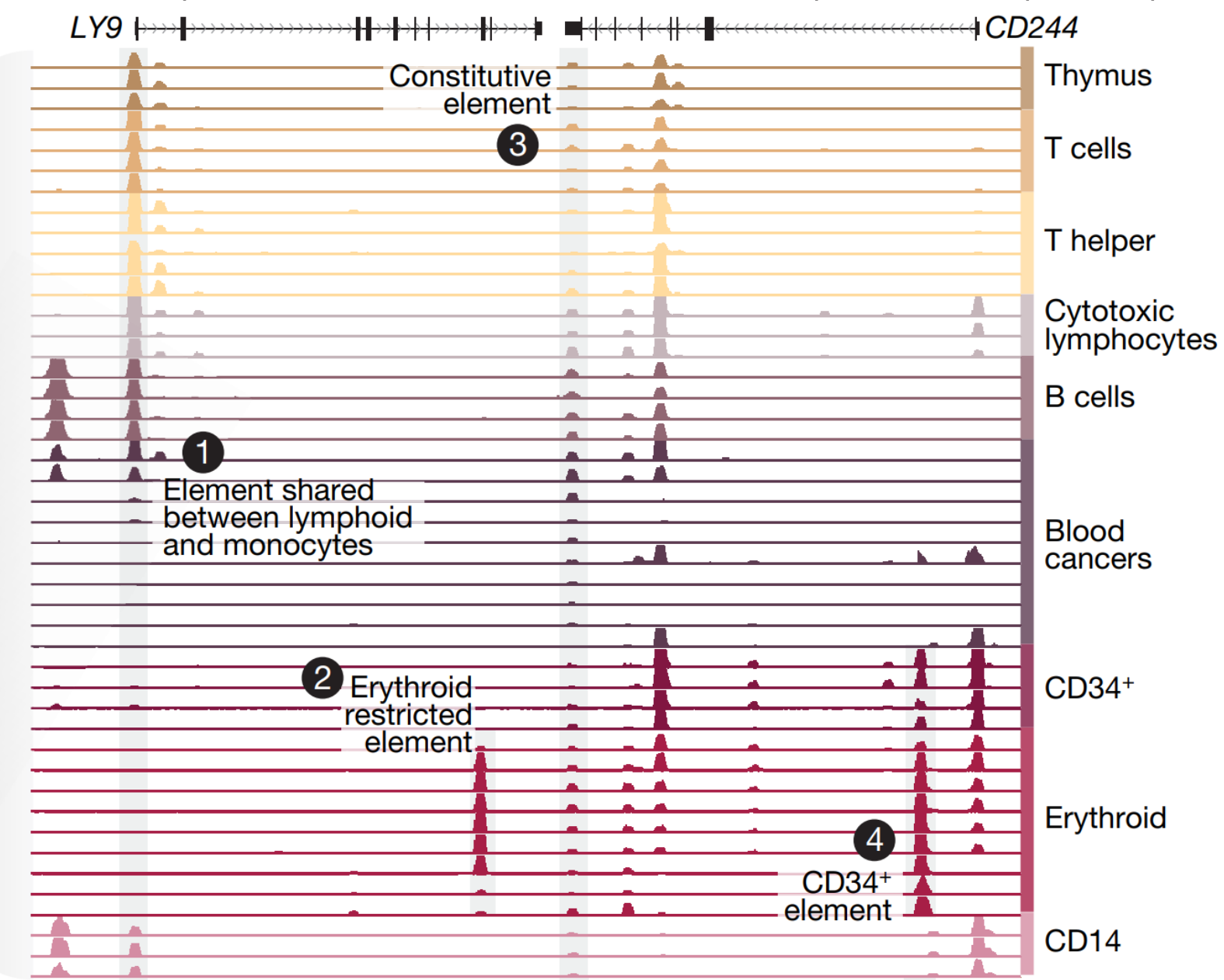


(B)



(C)





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TFBS extraction

inMOTIFin **New**LLM-extracted TF Targets **New**

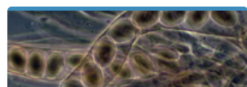
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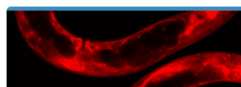
Browse JASPAR CORE for 6 different taxonomic groups



Fungi



Insecta



Nematoda



Plantae



Urochordata



Vertebrata

About JASPAR CORE

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The JASPAR CORE database contains a curated, non-redundant set of profiles, derived from published collections of experimentally defined transcription factor binding sites for eukaryotes. JASPAR is committed to providing open data access and non-redundant, high-quality binding profiles.

When should it be used? When seeking models for specific factors or structural classes, or if experimental evidence is paramount.



Q&A Forum

Ask questions about JASPAR here



RESTful API

Access JASPAR database programmatically



Download

Batch download PFMs, TFFMs, sites, and more



pyJASPAR

A Pythonic interface to JASPAR motifs

The high-quality transcription factor binding profile database

[Read more about JASPAR](#)[JASPAR interactive tour](#)

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[PubMed](#) | [NAR](#)

Baydar Ovek D, Rauluseviciute I, Aronsen DR, Blanc-Mathieu R, Bonthuis I, de Beukelaer H, Ferenc K, Jegou A, Kumar V, Lemma RB, Lucas J, Pochon M, Yun CM, Ramalingam V, Deshpande SS, Patel A, Marinov GK, Wang AT, Aguirre A, Castro-Mondragon JA, Baranasik D, Chèneby J, Gundersen S, Johansen M, Khan A, Kuijjer ML, Hovig E, Lenhard B, Sandelin A, Vandepoele K, Wasserman WW, Parcy F, Kundaje A, Mathelier A **JASPAR 2026: expansion of transcription factor binding profiles and integration of deep learning models** *Nucleic Acids Res.* 2026 Jan 6;54(D1):D184-D193.; doi: [10.1093/nar/gkaf1209](https://doi.org/10.1093/nar/gkaf1209)

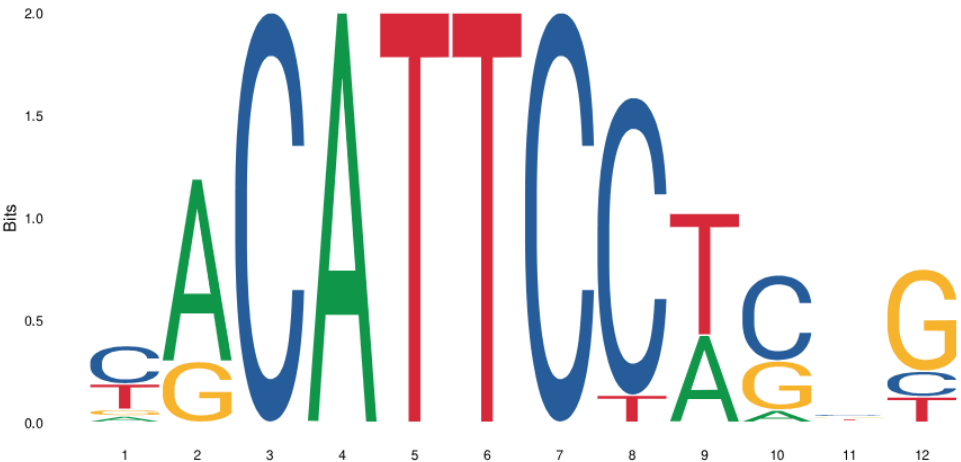
Profile summary

 Add

Name:	TEAD1
Matrix ID:	MA0090.1
Class:	TEA domain factors
Family:	TEF-1-related factors
Collection:	CORE
Taxon:	Vertebrates
Species:	Homo sapiens
Data Type:	COMPILED
Validation:	9571041
Uniprot ID:	P28347
Source:	
Comment:	

Sequence logo

 Download SVG



Frequency matrix

 JASPAR

 TRANSFAC

 MEME

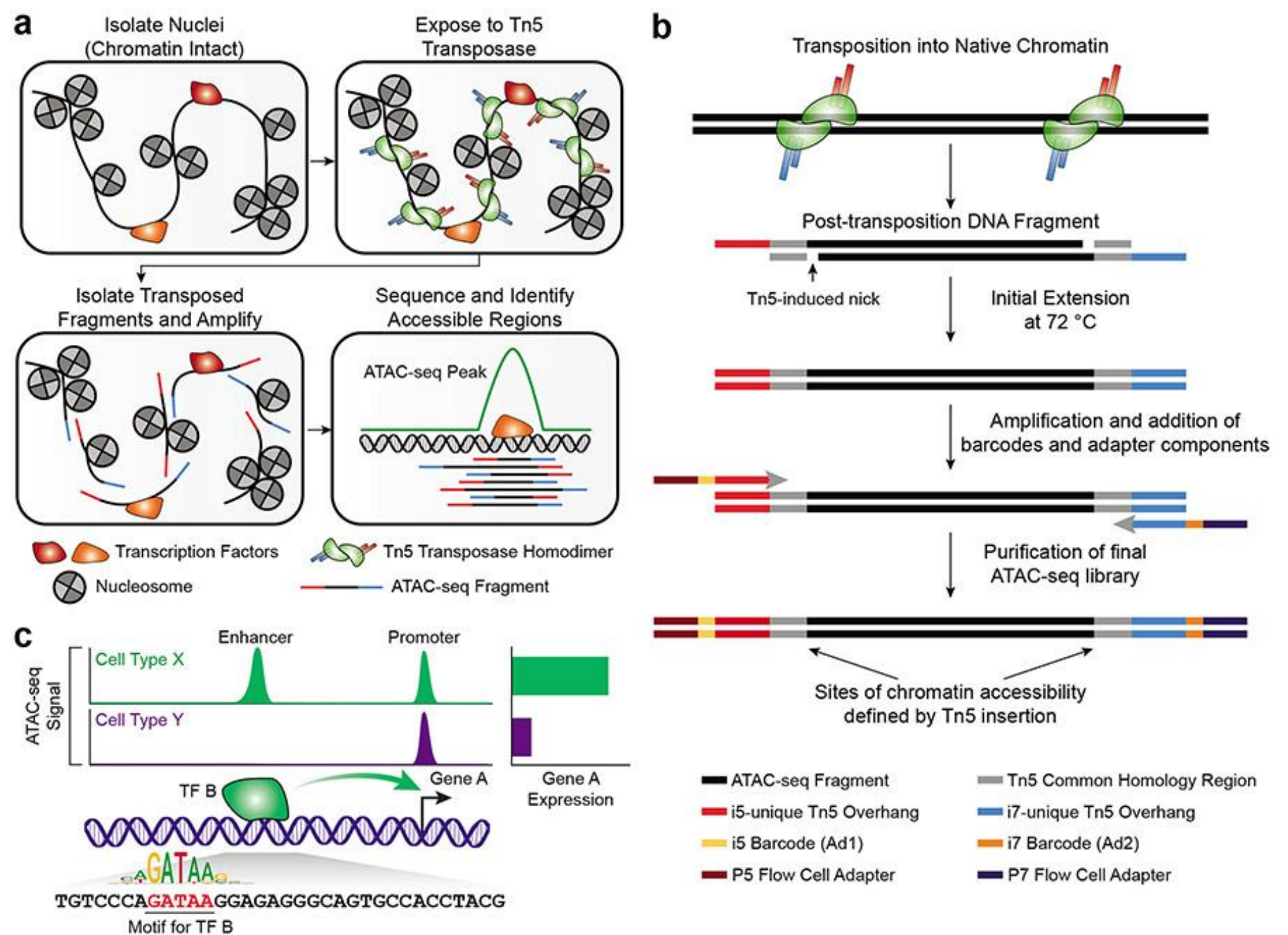
 RAW PFM

 Reverse comp.

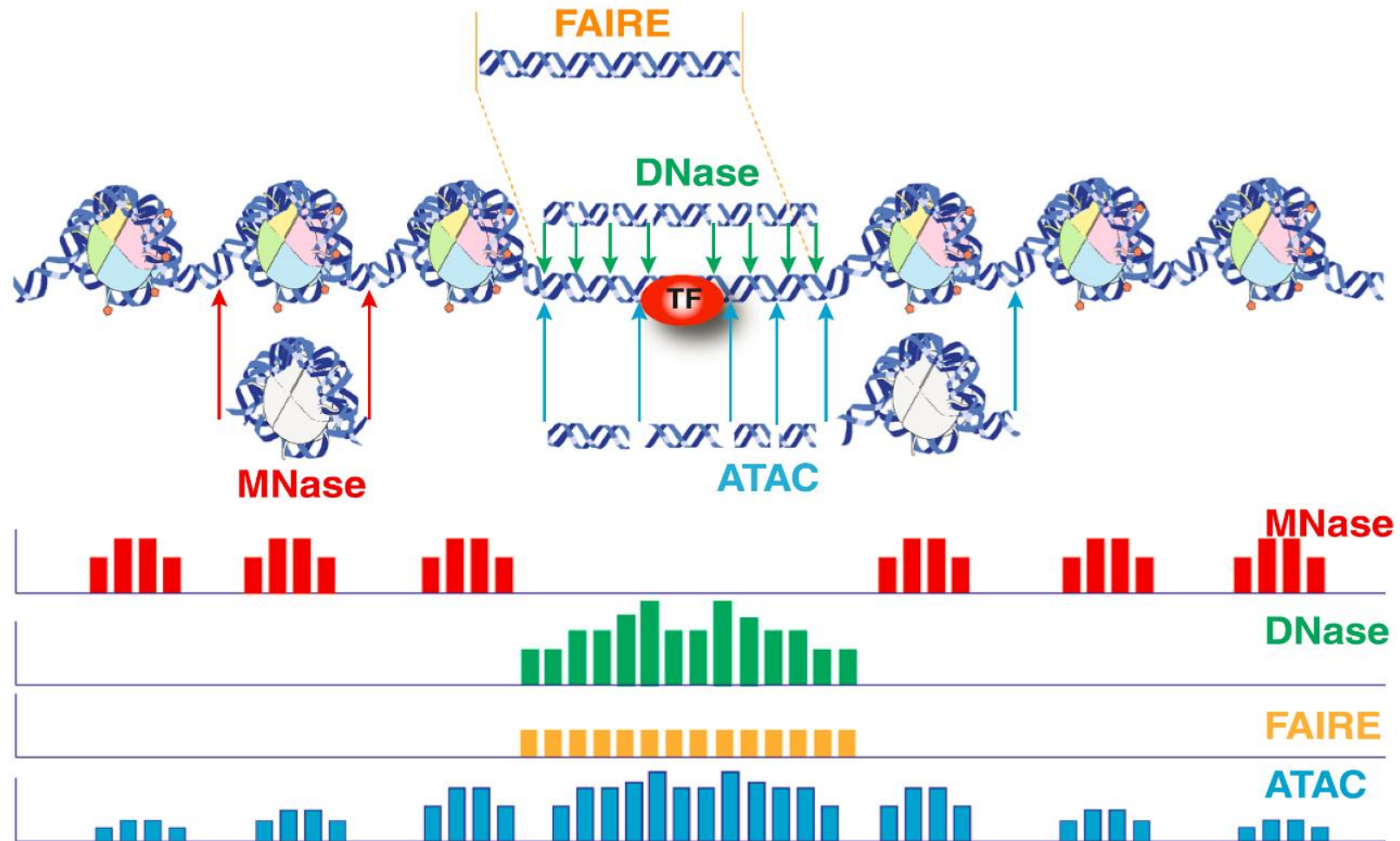
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C [	6	0	12	0	0	0	12	11	0	7	4	2	]
G [	1	3	0	0	0	0	0	0	0	4	3	8	]
T [	4	0	0	0	12	12	0	1	7	0	3	2	]

ATAC-seq

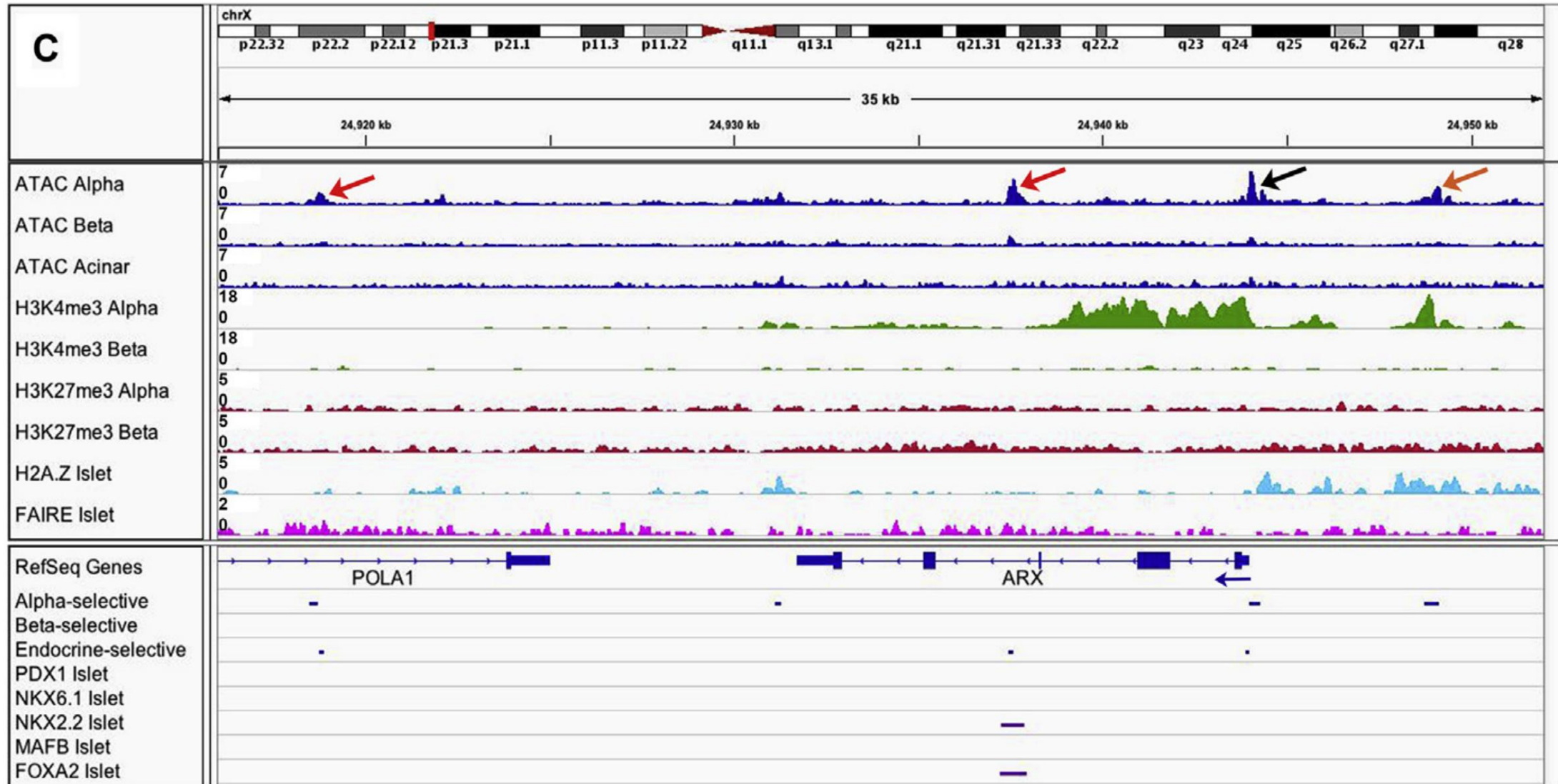
ATAC-seq



ATAC-seq



Reading the output of ATAC-seq



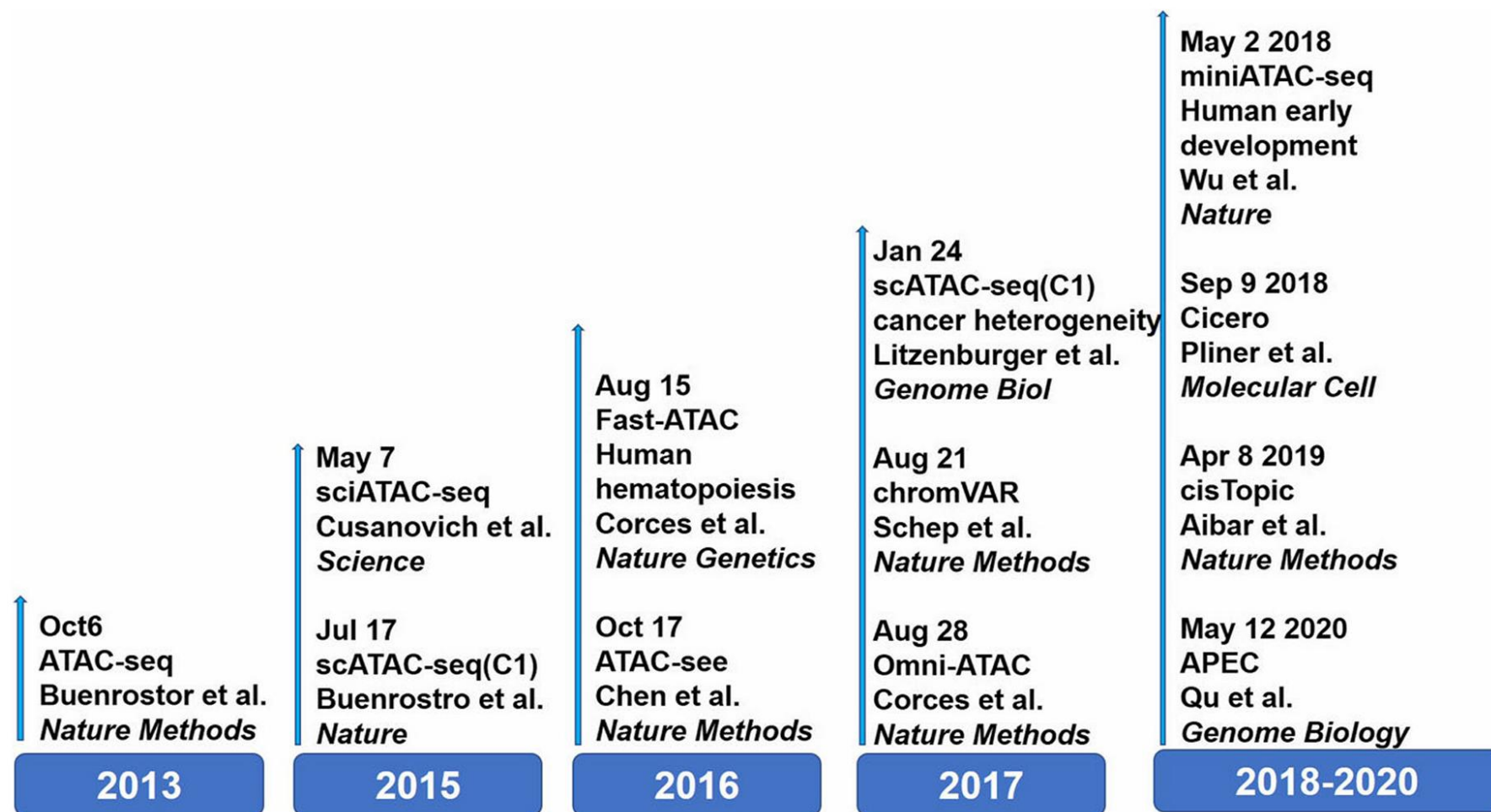


Fig. 3 Development of ATAC-seq and analysis tools

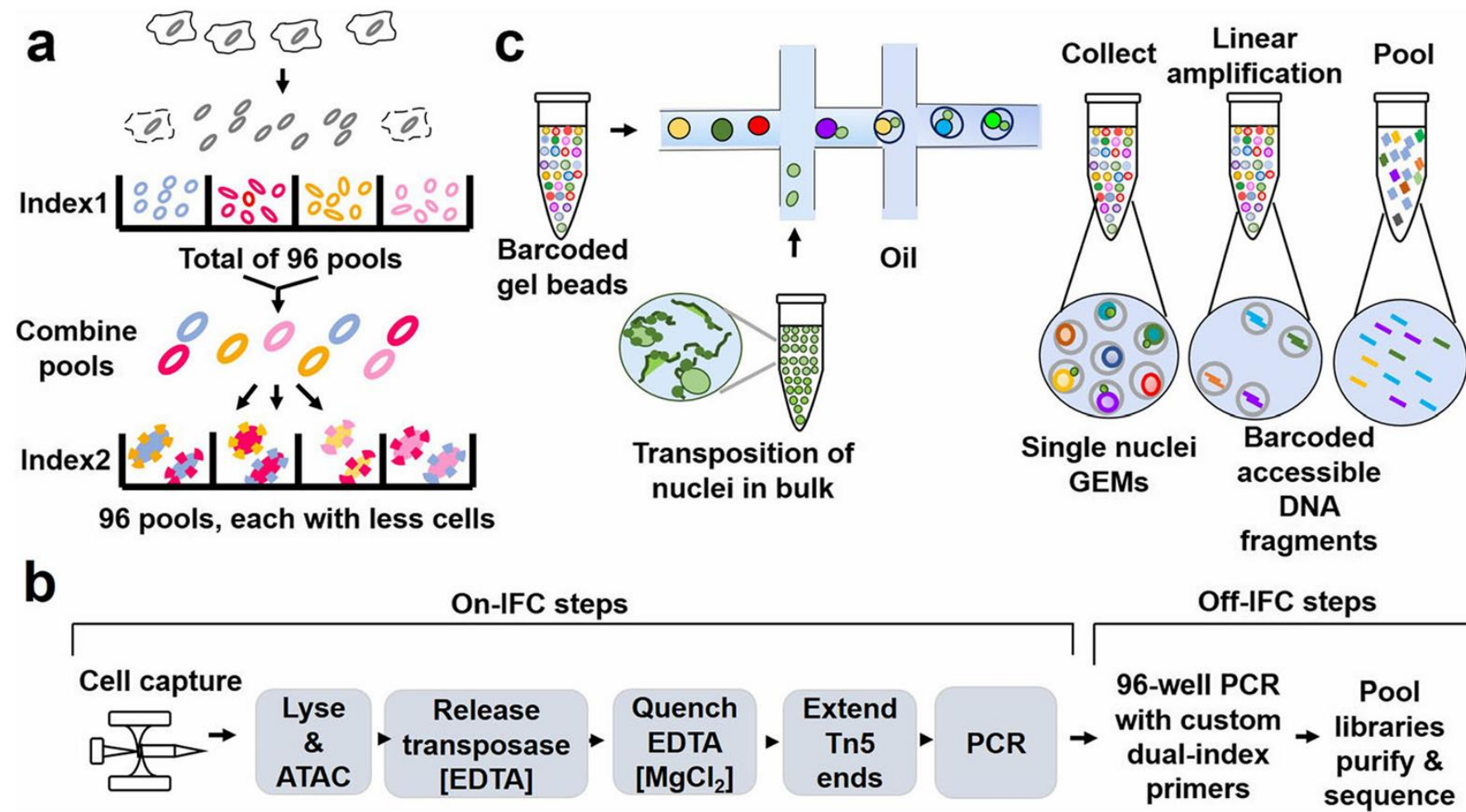


Fig. 4 Workflows of single-cell ATAC-seq. **a** The combinatorial indexing method of sci-ATAC-seq. The first barcodes are introduced by Tn5 transposase and the second indexing is introduced by amplification using primers containing a second barcode. **b** scATAC-seq based on the integrated fluidic circuit (IFC). In scATAC-seq using a microfluidics platform (Fluidigm), after transposition and PCR on the IFC, libraries were collected and PCR amplified with cell-identifying barcoded primers. Single-cell libraries were then pooled and sequenced on a high-throughput sequencing instrument. **c** Workflow of droplet-based scATAC-seq (10 × ATAC-seq) implemented on the Chromium platform (10 × Genomics). GEM: gel bead in emulsion

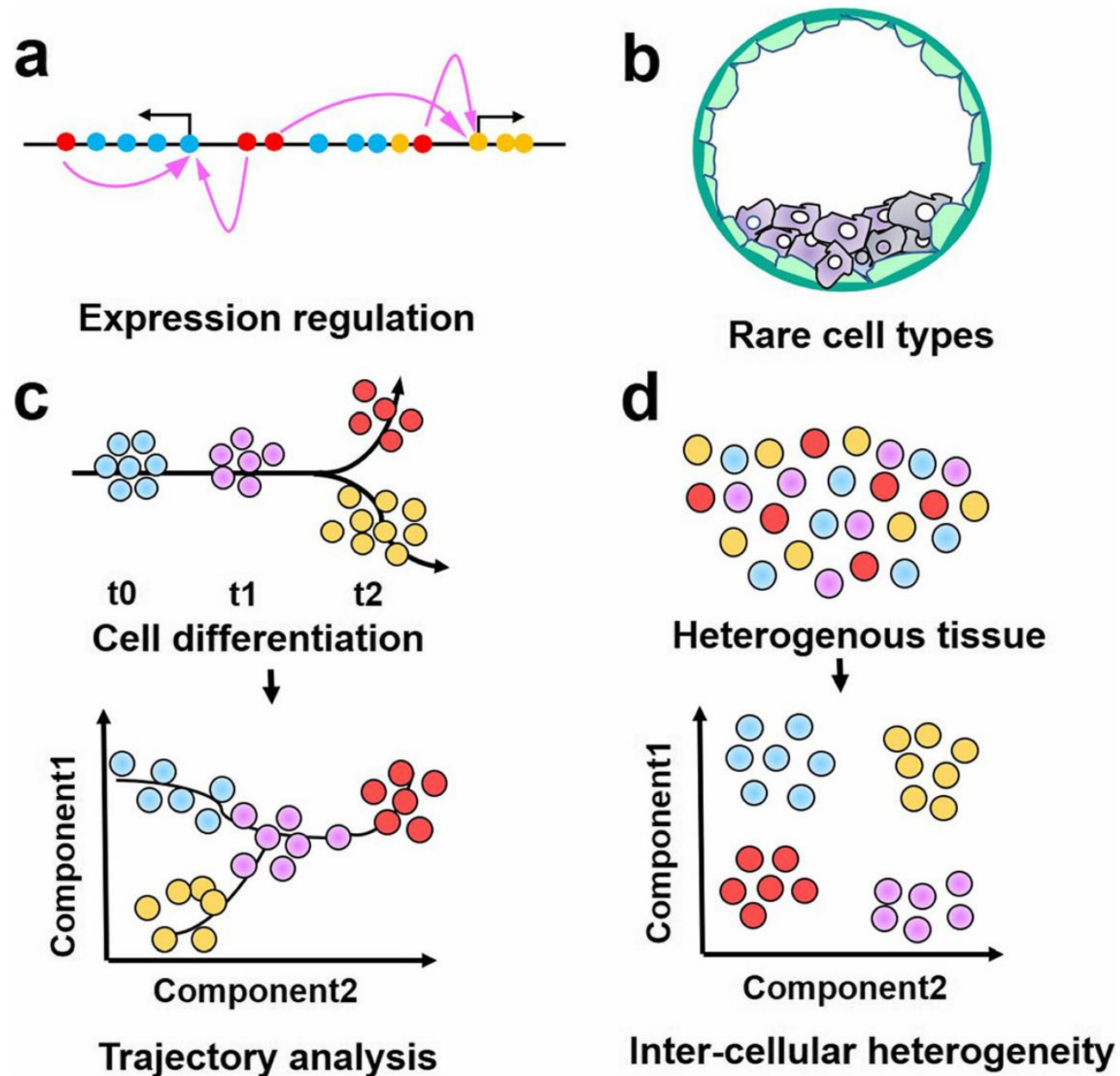
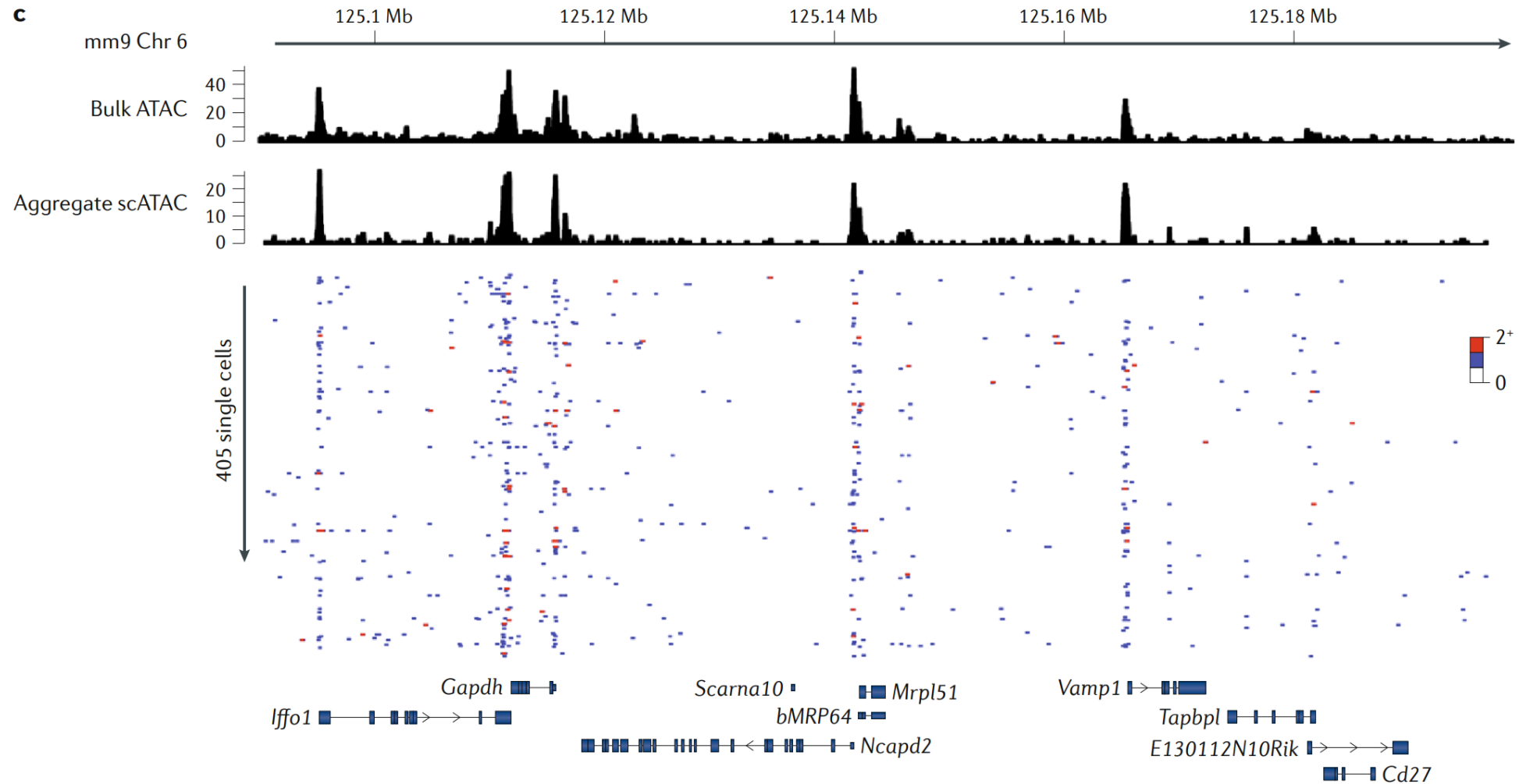


Fig. 5 Future applications of single-cell epigenomics. **a** Long-range interaction of regulatory elements. **b** Chromatin dynamics in rare cell types (e.g., early embryos). **c** Cell lineage tracking. **d** Deconvolution of intercellular heterogeneity

Population-scale measurements of chromatin accessibility reflect the average accessibility of a heterogeneous collection of single molecules



ChIP-seq for Histone modifications and TFBSs & ChromHMM

ChIP-seq overview

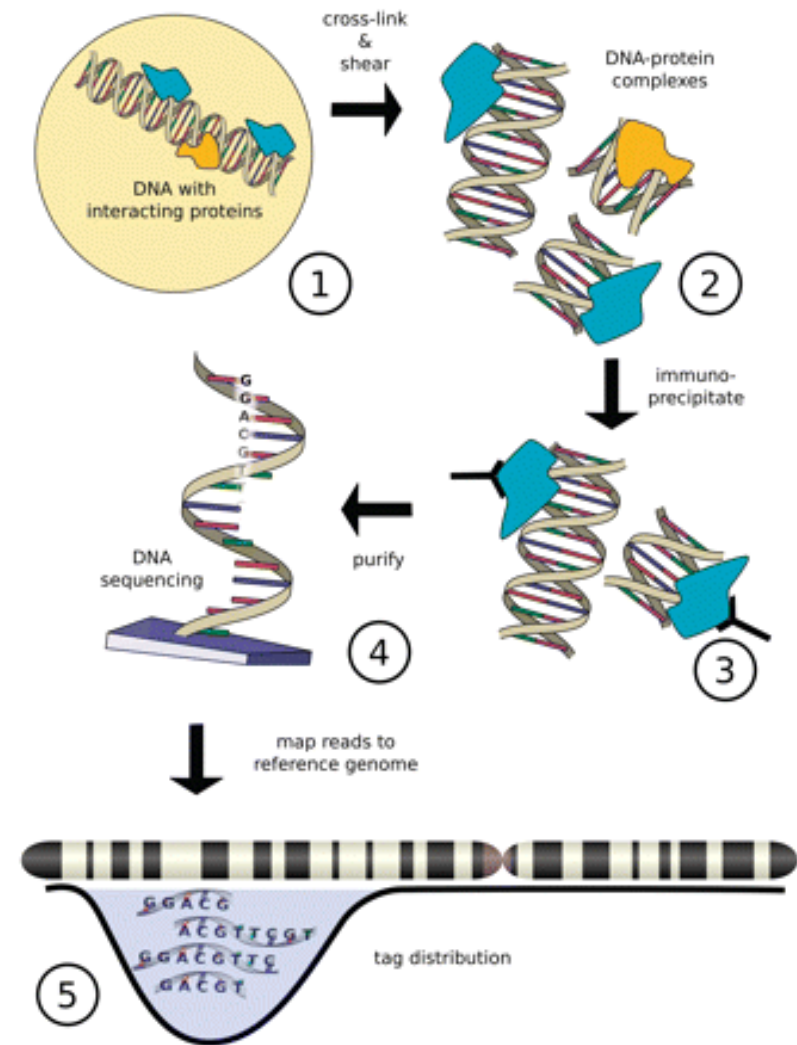
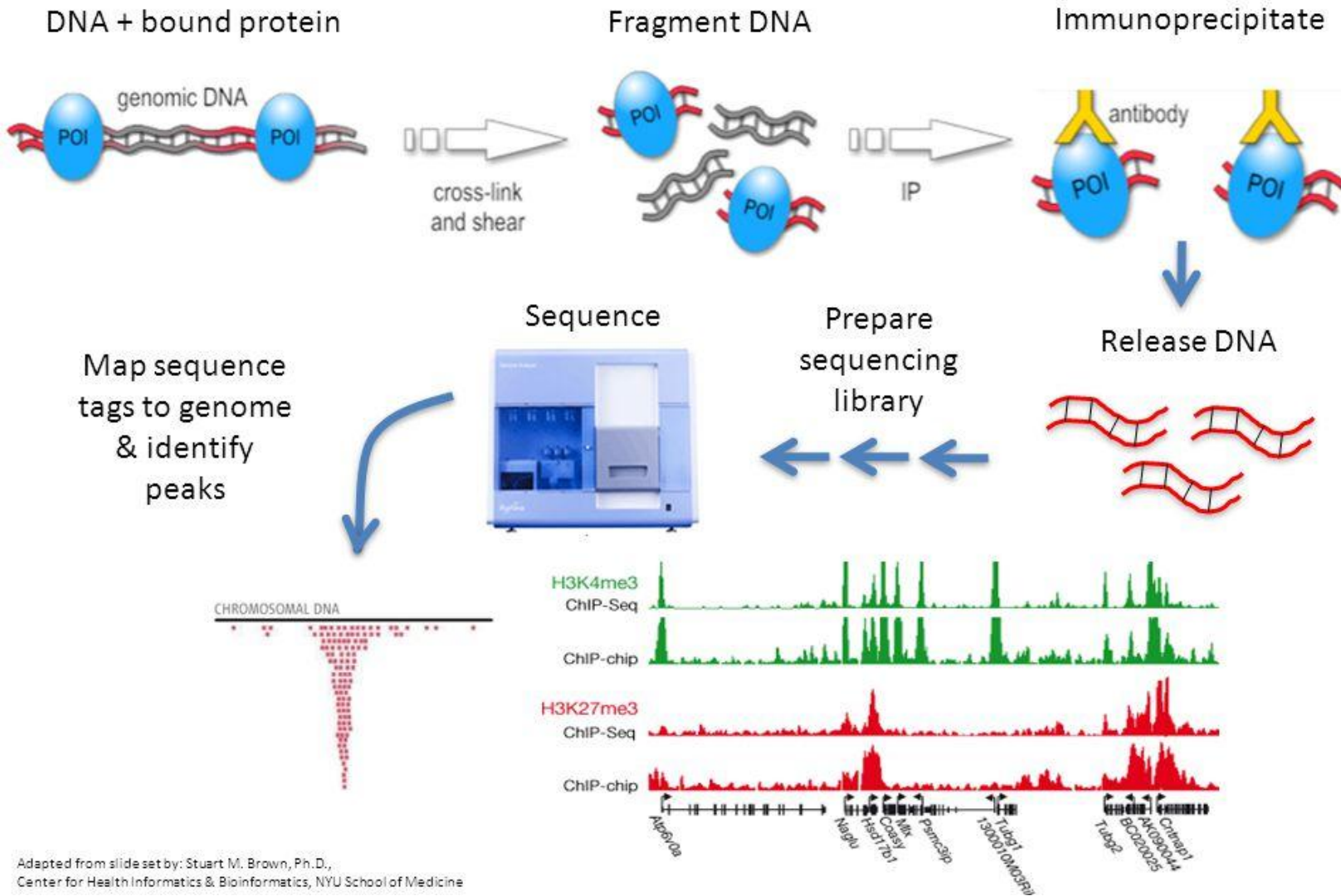
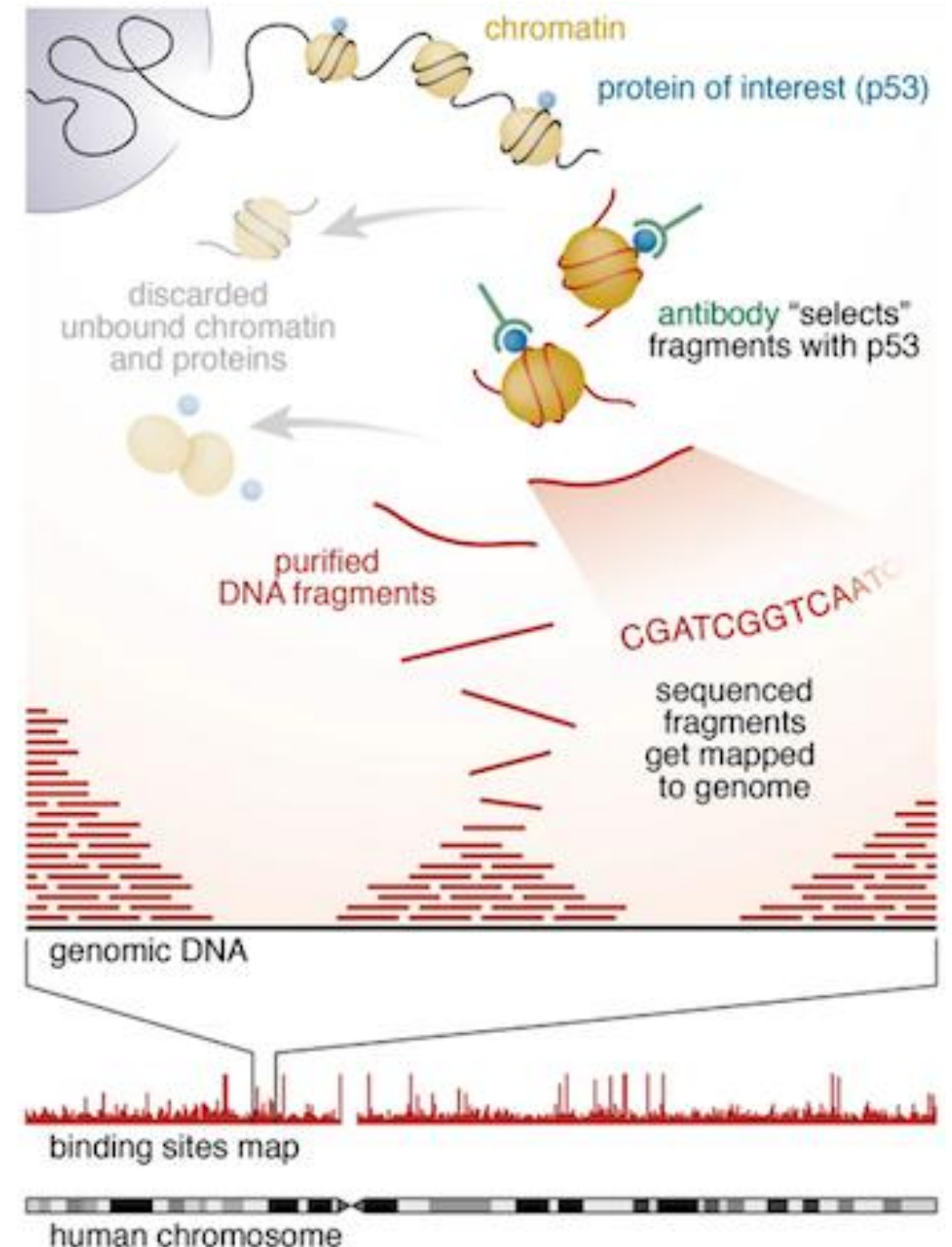
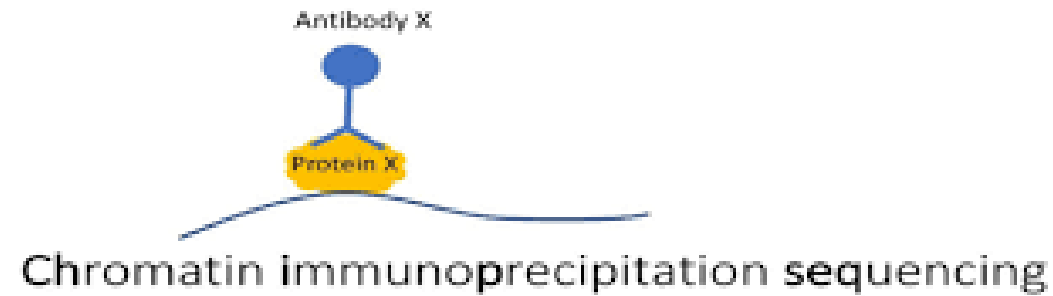
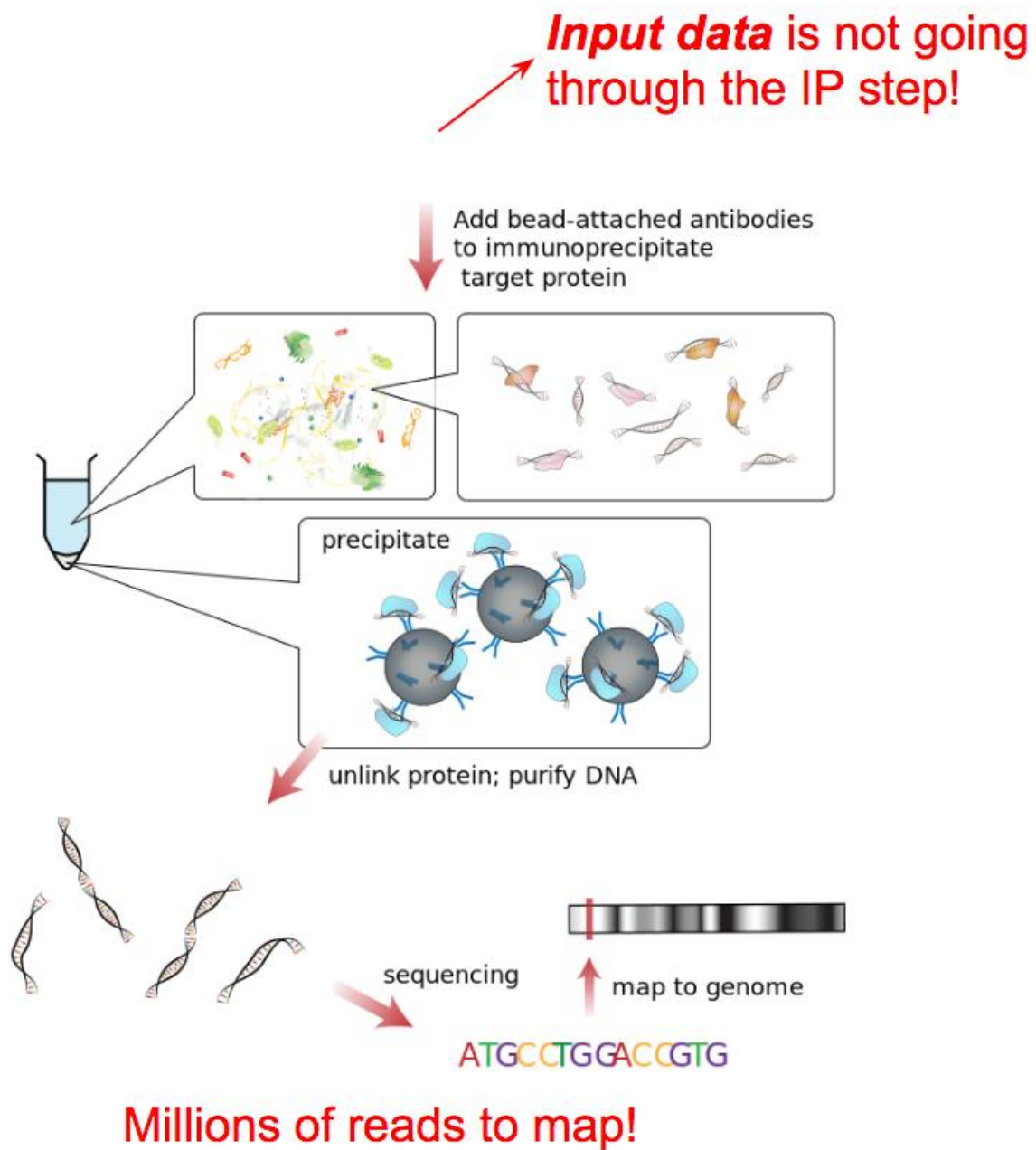
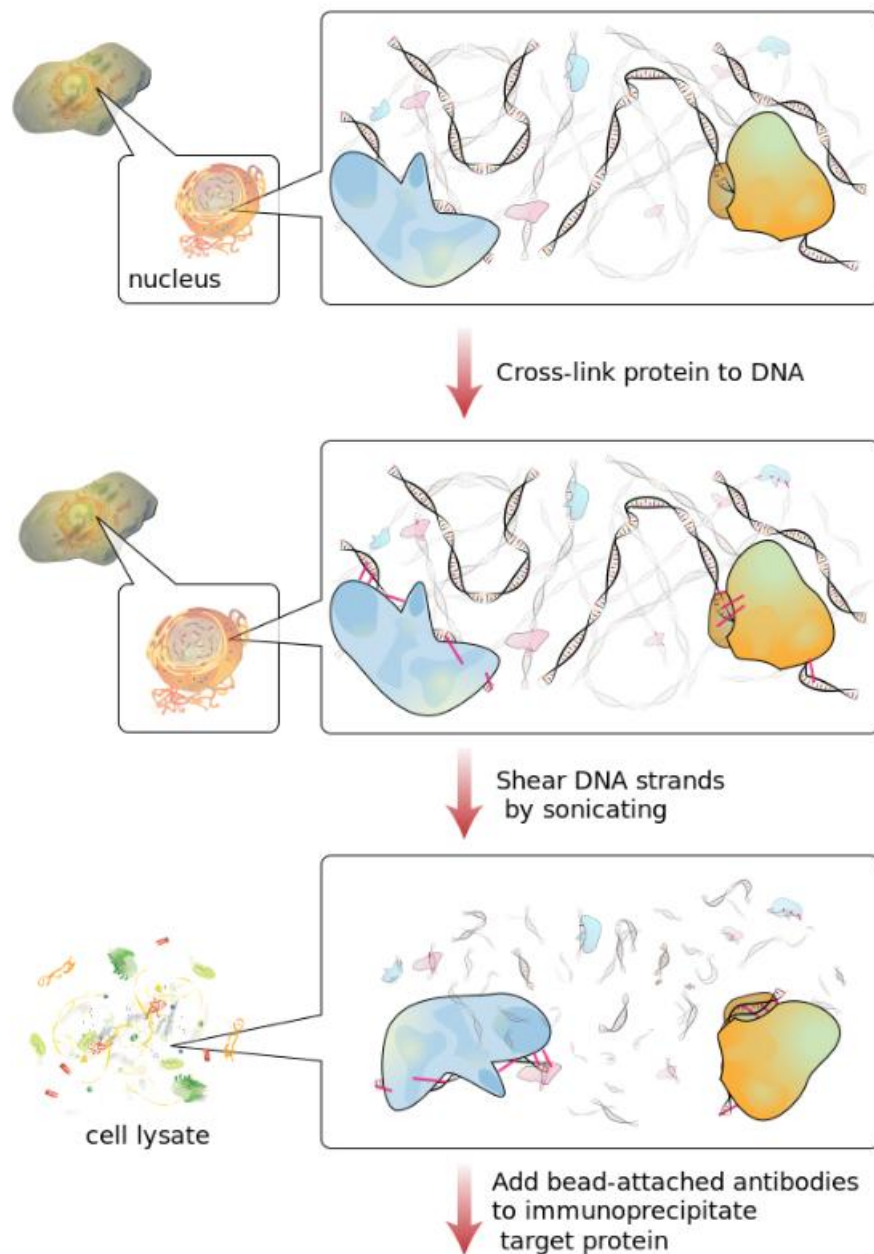


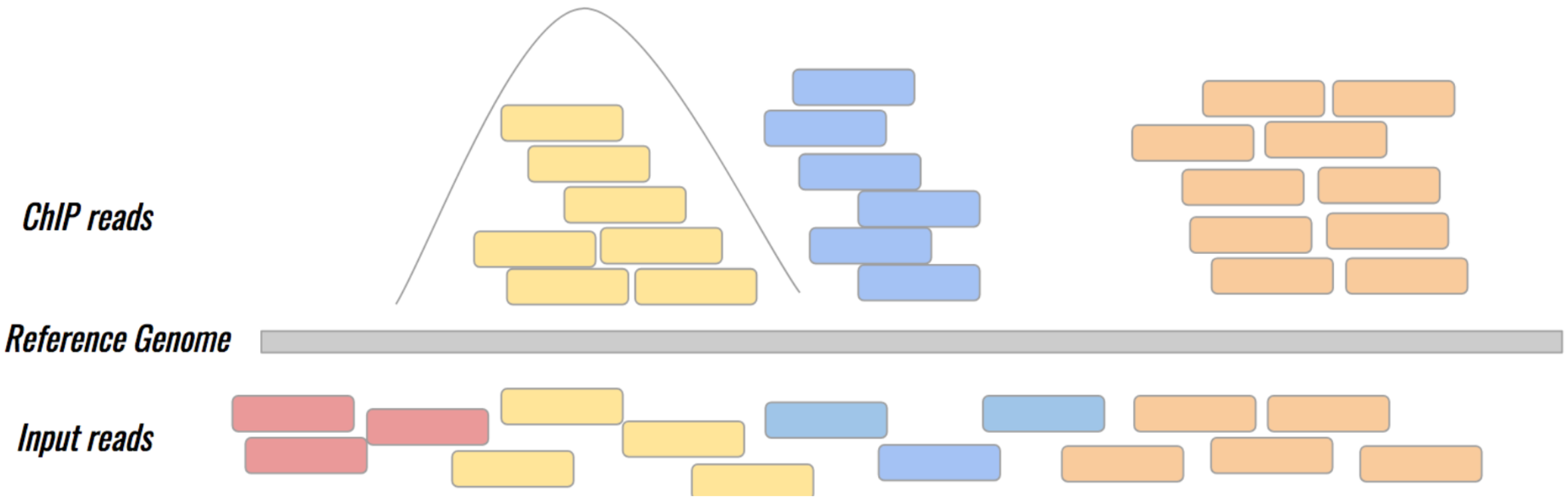
Figure from Szalkowski, A.M, and Schmid, C.D.(2010). *Rapid innovation in ChIP-seq peak-calling algorithms is outdistancing benchmarking efforts.* Briefings in Bioinformatics.

ChIP seq

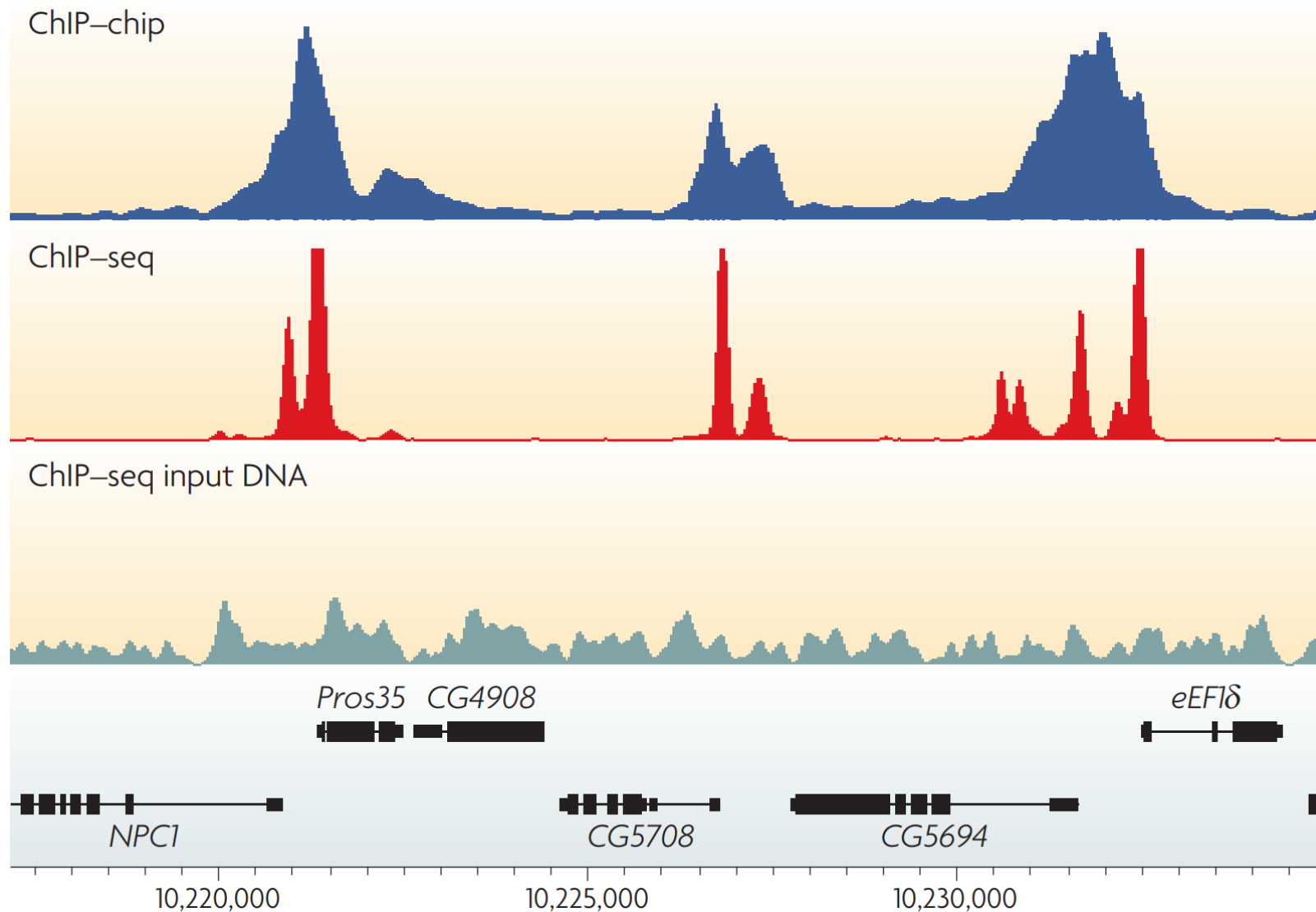




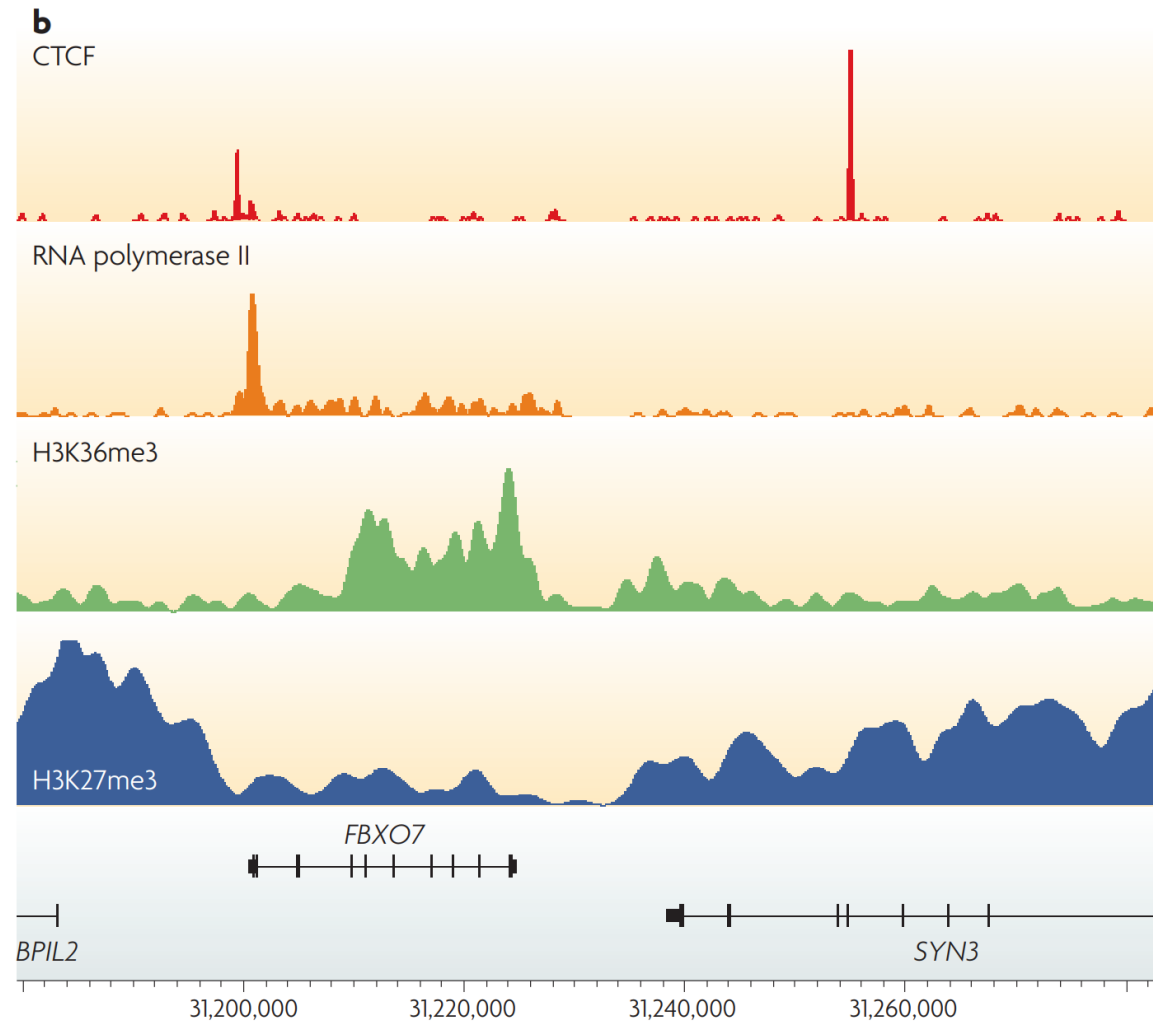
ChIP vs Input



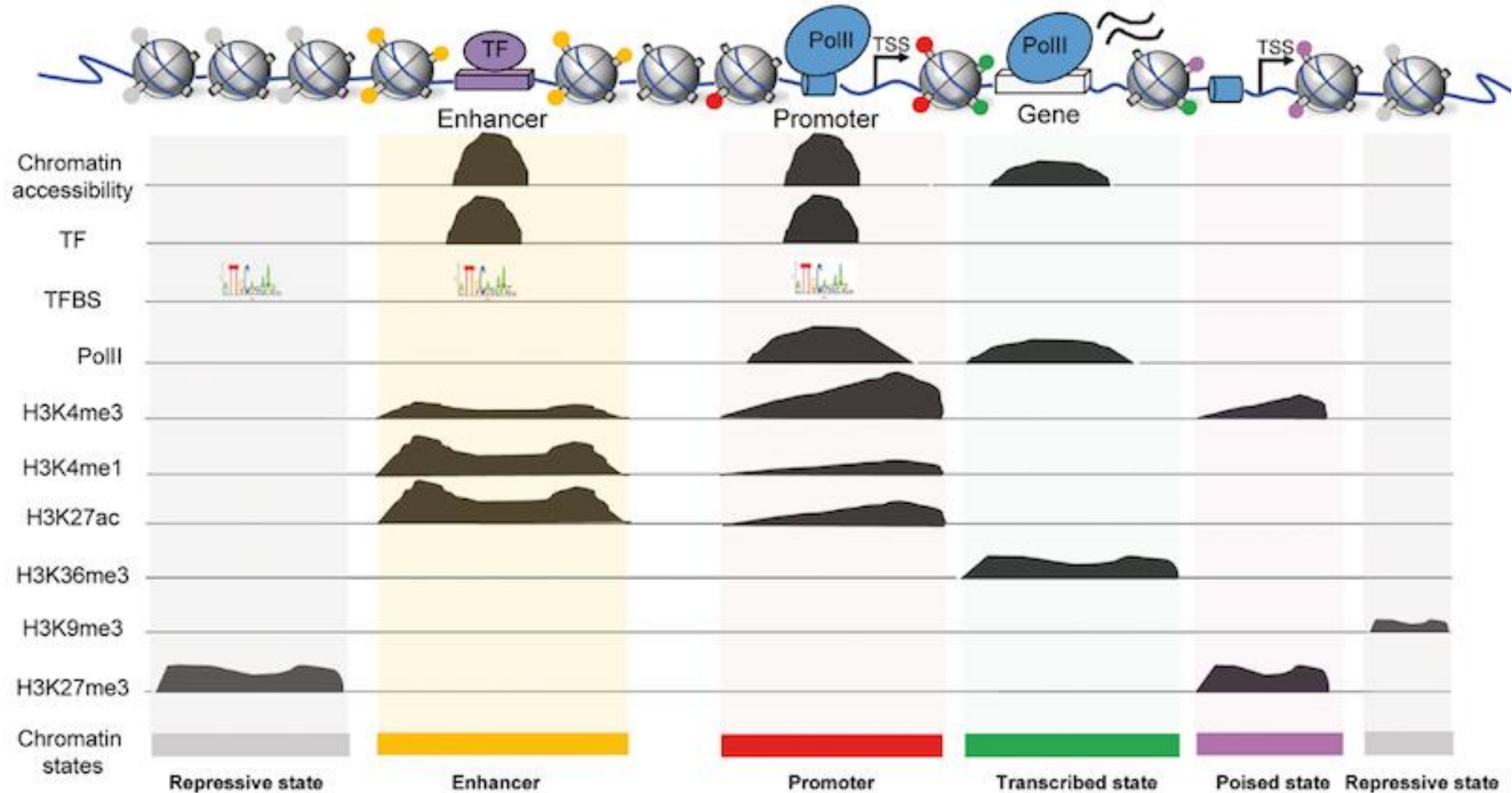
ChIP profiles: Examples of the profiles generated by chromatin immunoprecipitation



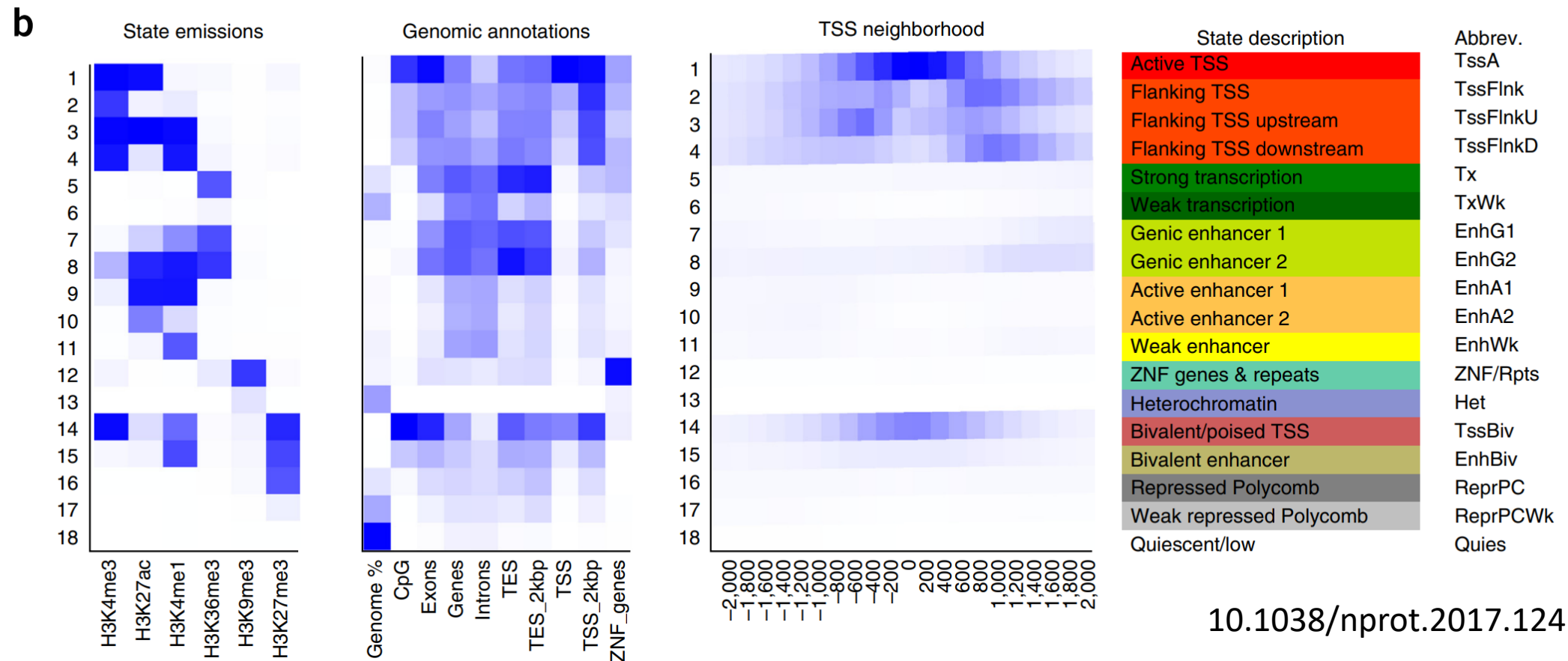
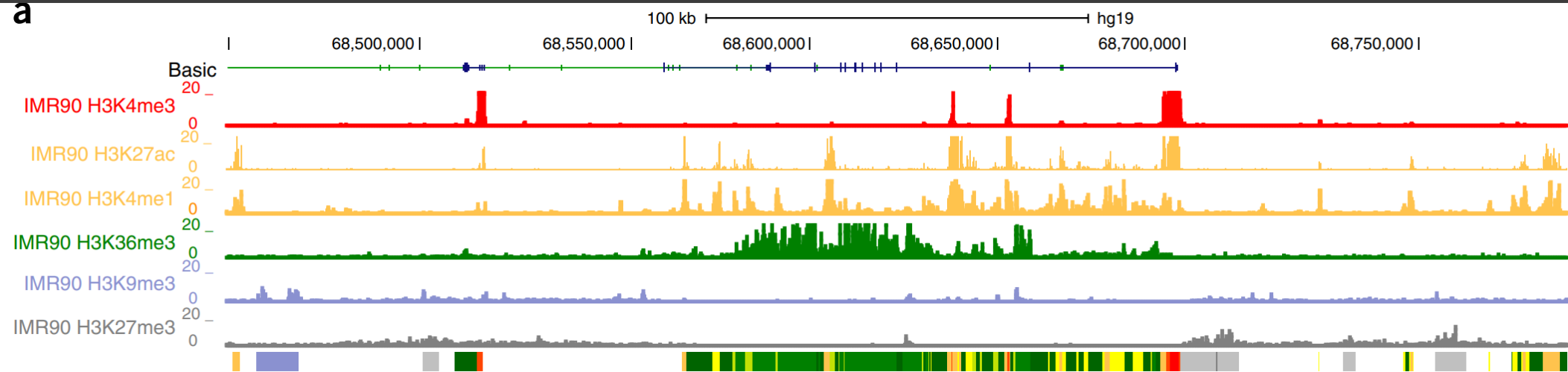
ChIP profiles: Examples of different types of ChIP–seq tag density profiles in human T cells



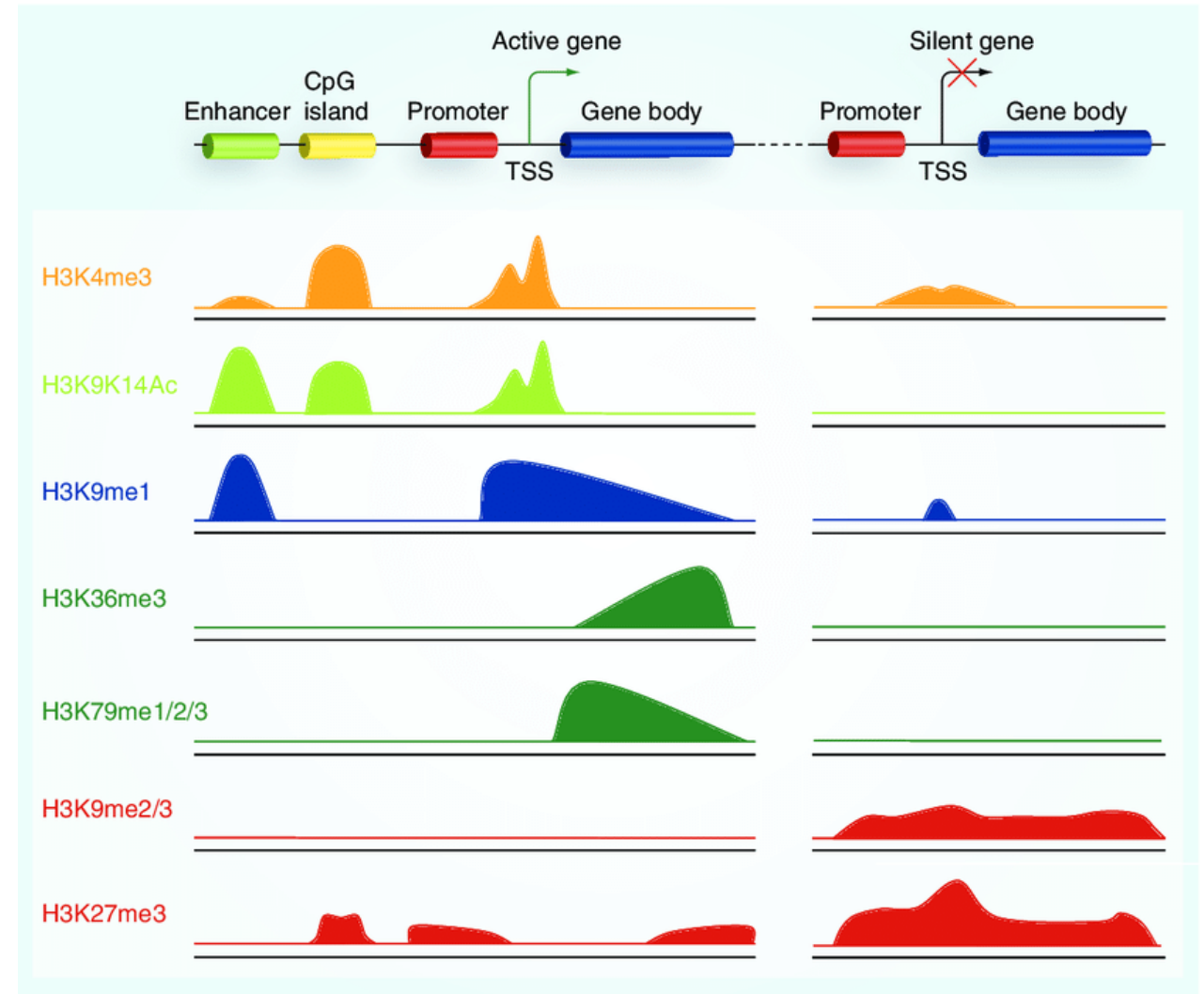
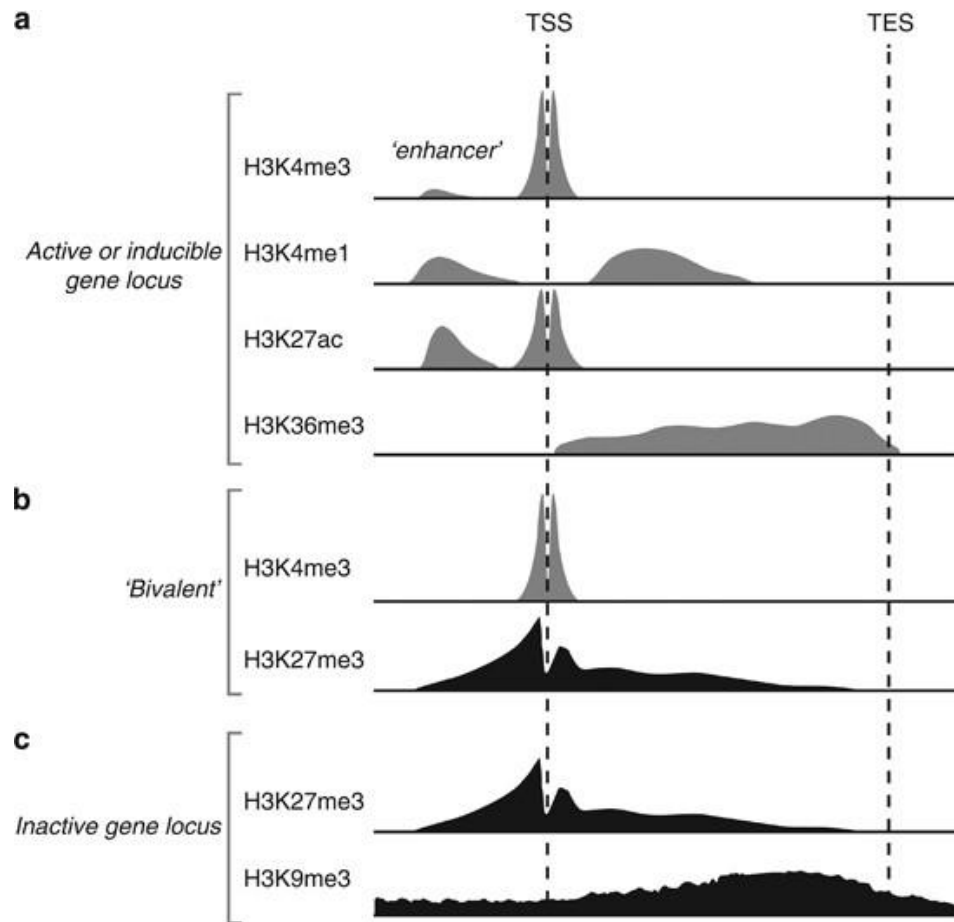
Types of ChIP-seq



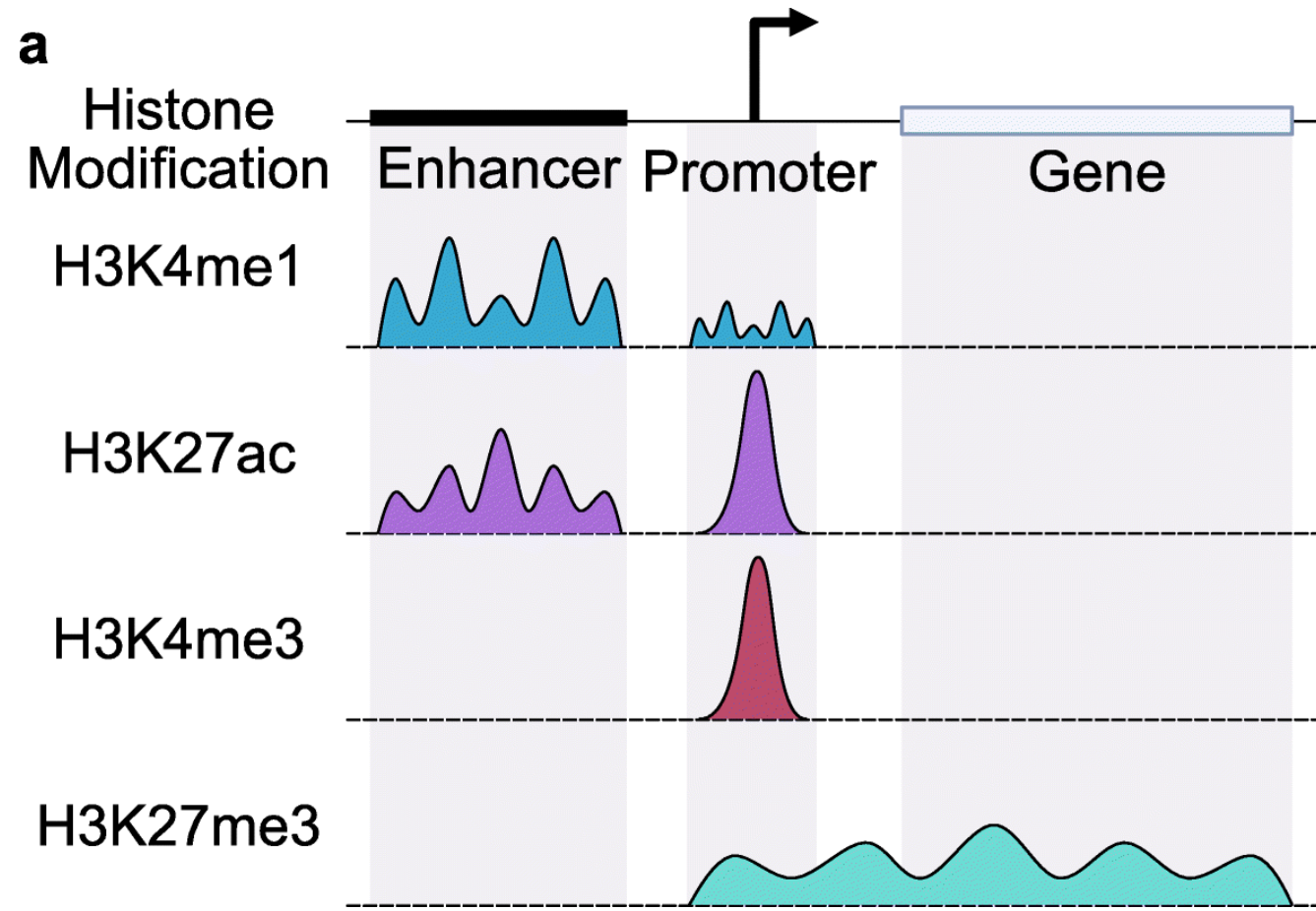
Chrom HMM

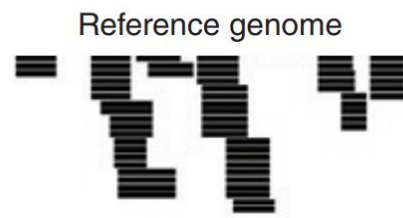


Peak types and locations of histone methylation



Peak types and locations of histone modifications

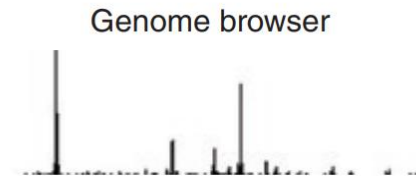




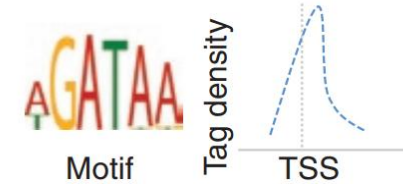
Reference genome

Sequence alignment

(no longer a bottleneck for data analysis)



Genome browser



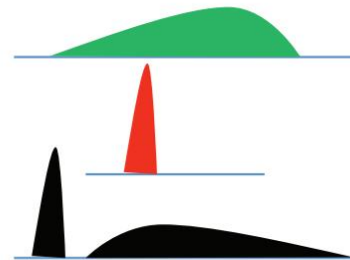
Motif

Tag density

TSS

Inspect data quality

- Check report from vendor-supplied analytical pipeline
- Visually inspect with a genome browser
- Identify motif in tag-enriched region
- Examine the profile at certain genomic features
- Confirm by quantitative PCR



Peak calling (choose the right tool)

Type of peak	Example	Representative tools
Broad	H3K27me3	CCAT, SICER
Sharp	CTCF	MACS
Sharp & broad	Pol II	ZINBA

Figure 2 Common procedures for ChIP-seq data analysis. After base-calling, short-read sequences are aligned to a reference genome. Data quality is assessed by a combination of various strategies, such as visual inspection with a genome browser, motif identification and confirmation by ChIP and quantitative PCR. The initial inspection or prior knowledge provides information about whether the peaks are broad or sharp or both. Various algorithms (bottom right) have been developed for the identification of peaks of these three groups.

RESEARCH

Open Access

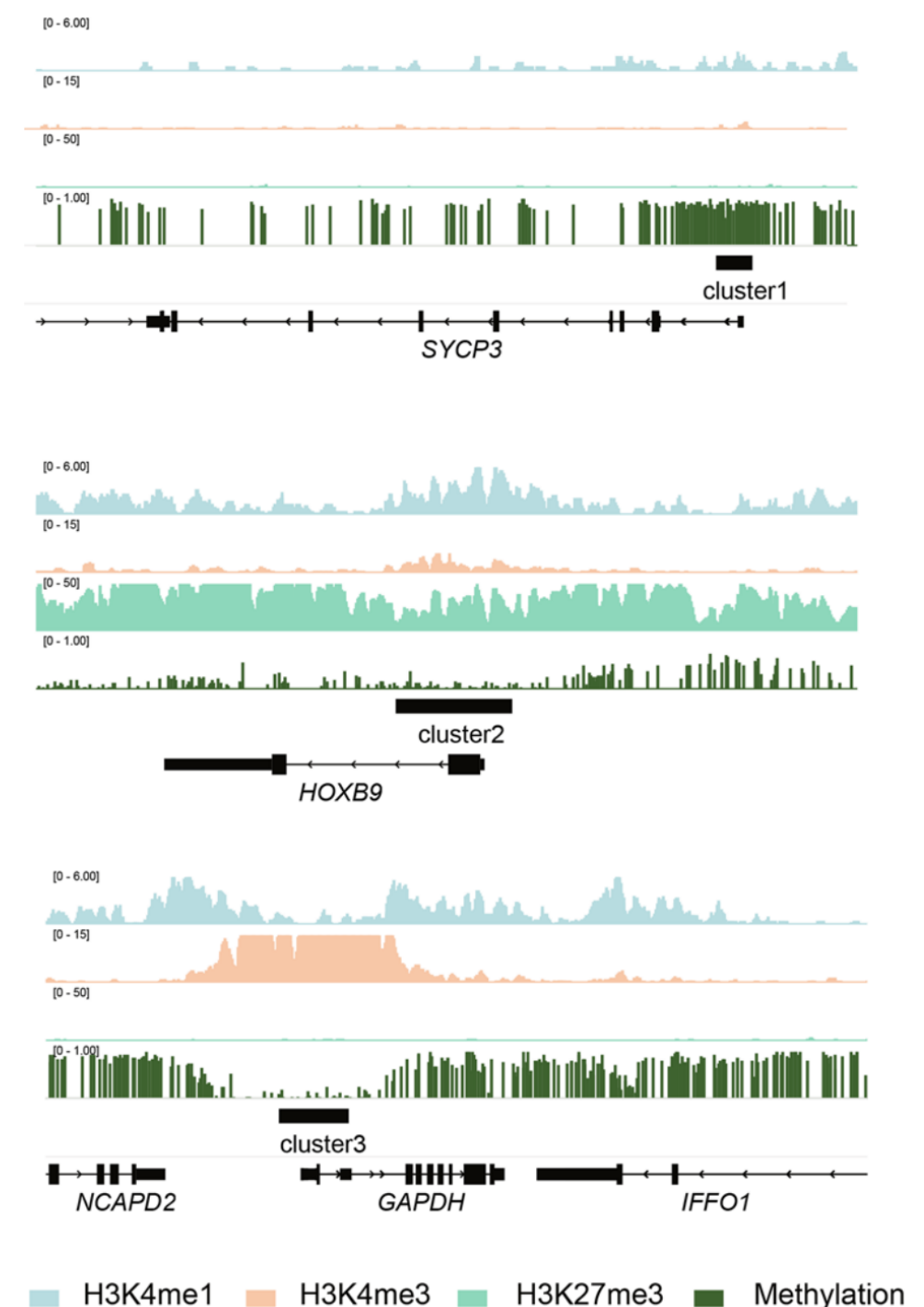
H3K27me3-H3K4me1 transition at bivalent promoters instructs lineage specification in development

Yue Yu^{1†}, Xinjie Li^{1†}, Rui Jiao², Yang Lu¹, Xuan Jiang^{1*} and Xin Li^{1,3*} 

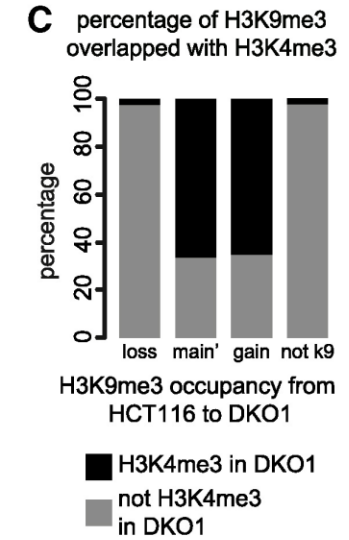
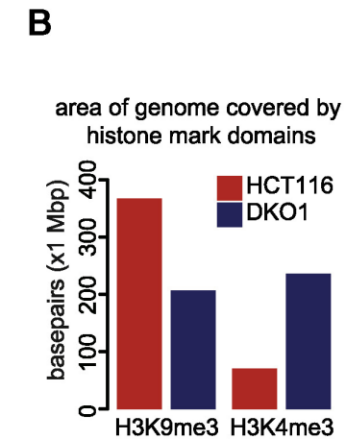
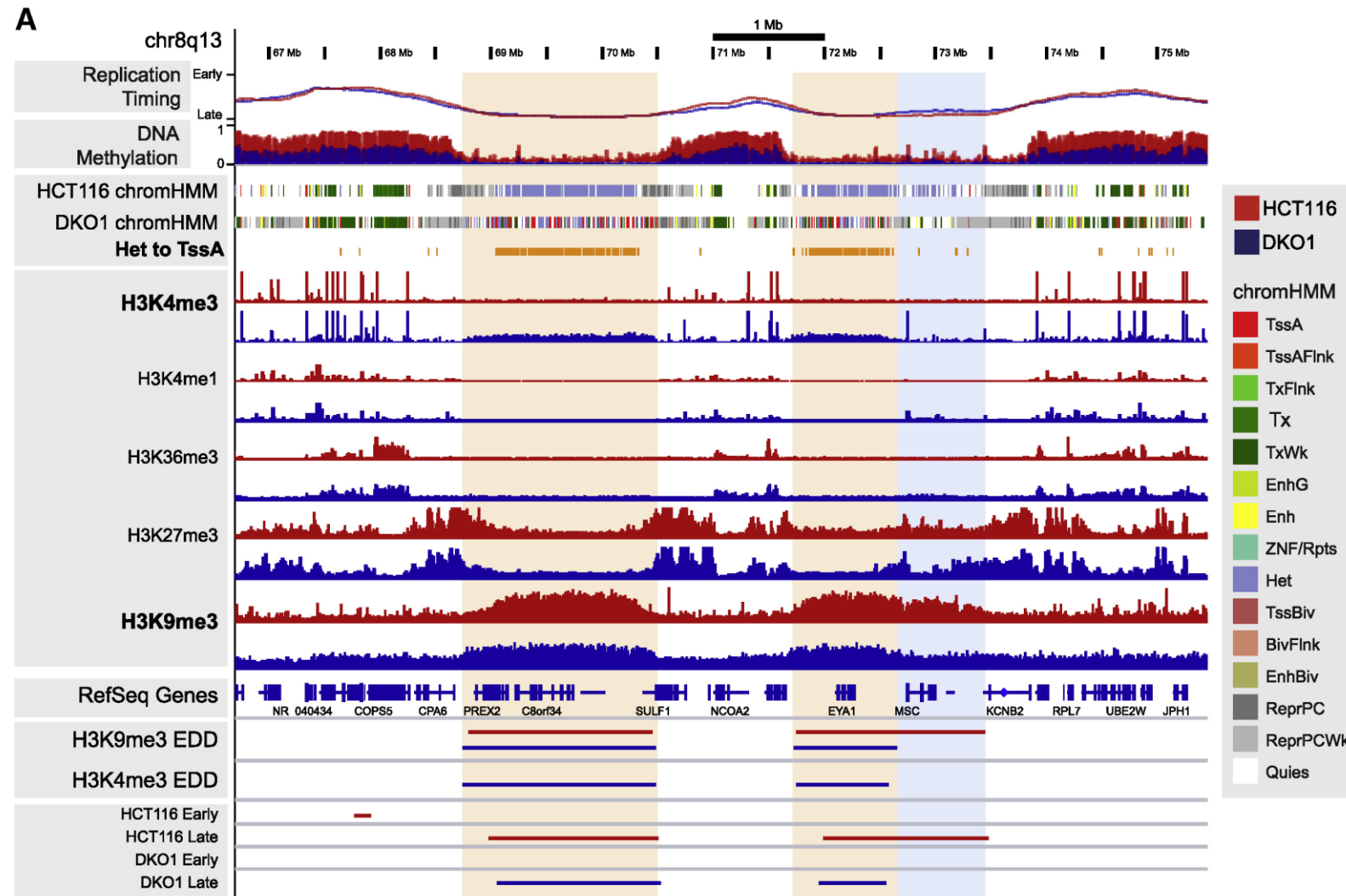
Abstract

Background Bivalent genes, of which promoters are marked by both H3K4me3 (trimethylation of histone H3 on lysine 4) and H3K27me3 (trimethylation of histone H3 on lysine 27), play critical roles in development and tumorigenesis. Monomethylation on lysine 4 of histone H3 (H3K4me1) is commonly associated with enhancers, but H3K4me1 is also present at promoter regions as an active bimodal or a repressed unimodal pattern. Whether the co-occurrence of H3K4me1 and bivalent marks at promoters plays regulatory role in development is largely unknown.

Results We report that in the process of lineage differentiation, bivalent promoters undergo H3K27me3-H3K4me1 transition, the loss of H3K27me3 accompanies by bimodal pattern loss or unimodal pattern enrichment of H3K4me1. More importantly, this transition regulates tissue-specific gene expression to orchestrate the development. Furthermore, knockout of *Eed* (Embryonic Ectoderm Development) or *Suz12* (Suppressor of Zeste 12) in mESCs (mouse embryonic stem cells), the core components of Polycomb repressive complex 2 (PRC2) which catalyzes H3K27 trimethylation, generates an artificial H3K27me3-H3K4me1 transition at partial bivalent promoters, which leads to up-regulation of meso-endoderm related genes and down-regulation of ectoderm related genes, thus could explain the observed neural ectoderm differentiation failure upon retinoic acid (RA) induction. Finally, we find that lysine-specific demethylase 1 (LSD1) interacts with PRC2 and contributes to the H3K27me3-H3K4me1 transition in mESCs.



Reading the output of ChIP-Seq of Histone Modifications



Du Q, Smith G, Luu PL *et al.* DNA methylation is required to maintain both DNA replication timing precision and 3D genome organization integrity. Cell Reports